

Verification: $AI = IA = A$ for Square Matrices

Following Axler, *Linear Algebra Done Right 4th Ed.*, §3C–3D

Claim. *Let A be an $n \times n$ matrix and let I be the $n \times n$ identity matrix. Then $AI = IA = A$.*

We recall Theorem 3.51 from §3C: for two matrices C (of size $m \times c$) and R (of size $c \times n$), the k th column of the product CR is the linear combination of the columns of C with coefficients taken from the k th column of R :

$$(CR)_{\cdot, k} = R_{1, k} C_{\cdot, 1} + R_{2, k} C_{\cdot, 2} + \cdots + R_{c, k} C_{\cdot, c} = \sum_{j=1}^c R_{j, k} C_{\cdot, j}.$$

Also recall that the k th column of the identity matrix I is the vector e_k , which has a 1 in row k and 0's everywhere else:

$$I_{\cdot, k} = e_k, \quad \text{so} \quad I_{j, k} = \begin{cases} 1 & \text{if } j = k, \\ 0 & \text{if } j \neq k. \end{cases}$$

Part 1: $AI = A$.

Fix any column index $k \in \{1, \dots, n\}$. Applying Theorem 3.51(a) with $C = A$ and $R = I$:

$$(AI)_{\cdot, k} = \sum_{j=1}^n I_{j, k} A_{\cdot, j}.$$

Since $I_{j, k} = 0$ for $j \neq k$ and $I_{k, k} = 1$, every term in the sum vanishes except the $j = k$ term:

$$(AI)_{\cdot, k} = 1 \cdot A_{\cdot, k} = A_{\cdot, k}.$$

Because the k th column of AI equals the k th column of A for every k , we conclude $AI = A$. $\square$

Part 2: $IA = A$.

Fix any column index $k \in \{1, \dots, n\}$. Applying Theorem 3.51(a) with $C = I$ and $R = A$:

$$(IA)_{\cdot, k} = \sum_{j=1}^n A_{j, k} I_{\cdot, j} = A_{1, k} e_1 + A_{2, k} e_2 + \cdots + A_{n, k} e_n.$$

Expanding each scaled standard basis vector, the i th entry of this sum is

$$[(IA)_{\cdot,k}]_i = \sum_{j=1}^n A_{j,k} (e_j)_i = A_{i,k},$$

since $(e_j)_i = 1$ only when $j = i$ and is 0 otherwise. Hence $(IA)_{\cdot,k} = A_{\cdot,k}$ for every k , giving $IA = A$. $\square$

Conclusion. Both $AI = A$ and $IA = A$, so $AI = IA = A$.